Structure-Aware Representation Fusion for Improving CLIP Transfer on the LITE-Benchmark

1 Project Scope and Milestone 2 Focus

Our project studies transfer learning for low-resource technical image classification, with a focus on the circuit diagram subset of the LITE benchmark. In Milestone 1, we proposed **Structure-Aware Representation Fusion (SARF)**, a lightweight method that augments frozen CLIP image representations with structural edge-based features. For Milestone 2, we focus on the **data, preprocessing pipeline, experimental setup, and baseline evaluation** needed to support this direction. In addition to the required baselines, we also implemented a series of **SARF-based experiments**, ranging from **SARF-Sum** to **ensemble-based fusion**, in order to study how different structure-aware fusion strategies affect performance in this low-resource setting.

The main motivation is that circuit diagrams differ substantially from natural images: they contain sparse texture, repeated symbols, and heavy dependence on topology and connectivity. Because CLIP is pretrained primarily on natural-image data, we investigate whether incorporating structural cues improves transfer to this low-resource domain. In this milestone, we compare **standard baselines** against our **proposed SARF methods** in a controlled frozen-backbone setting.

2 Data

We use the **circuit diagram classification** task from the LITE benchmark [1]. This dataset contains **32 circuit function classes**, such as audio amplifier, power supply, timer, alarm, and sensor-related circuits. The benchmark is explicitly designed for low-resource vision settings and includes highly specialized technical images rather than natural photographs.

For our experiments, we use the provided train/validation/test split:

- **Training set:** 154 images
- **Validation set:** 100 images
- **Test set:** 1078 images

Each example consists of a circuit image and a class label indicating circuit function. A label map file is used to convert class IDs into human-readable class names. This setup follows the low-resource nature of LITE, where the training set is intentionally small and the test set is much larger, making generalization difficult. The small number of training examples relative to the number of classes makes this a challenging classification problem and motivates careful baseline selection.

3 Data Preprocessing

Our preprocessing pipeline is designed to support both standard RGB inputs and structure-enhanced edge inputs.

3.1 Dataset Loading and Label Mapping

We organize the dataset using a base directory containing the circuit images and metadata files. Specifically, the image files are stored in a dedicated `data/` folder, while the dataset splits are defined through separate CSV files for training, validation, and testing. We use four metadata files: `train.csv`, `val.csv`, `test.csv`, and

`label_map.csv`. The first three files specify the image names and corresponding class IDs for each split, while `label_map.csv` provides the mapping from numeric class IDs to human-readable class names.

We then construct the full image path for each example by joining the image filename from the CSV with the root image directory. This produces a new `image_path` field for each sample, which is later used for feature extraction and preprocessing.

Finally, we build a label dictionary from `label_map.csv` to map each class ID to its corresponding class name. Using this mapping, we create an ordered list of class labels and compute the total number of classes in the dataset.

3.2 RGB Pipeline

For the RGB branch, each image is loaded from disk, converted to RGB, and processed using the `openai/clip-vit-base-patch32` processor. The processor resizes and normalizes images according to the pretrained CLIP vision model requirements. No additional task-specific fine-tuning is applied to the vision backbone. This branch preserves the original appearance of the circuit image and serves as the main visual baseline.

3.3 Edge Pipeline

To inject structural information, we generate an edge-based representation for each circuit image. The preprocessing steps are:

1. Load the original image and convert it to RGB.
2. Optionally crop surrounding whitespace to reduce irrelevant background.
3. Convert the cropped image to grayscale.
4. Apply histogram equalization to enhance contrast.
5. Apply Gaussian blur to reduce noise.
6. Apply Canny edge detection with thresholds of 50 and 150.
7. Apply morphological closing and one-step dilation to strengthen thin circuit traces.
8. Convert the resulting edge map back to a three-channel RGB image so it can be processed by the CLIP image encoder.

This edge pipeline is intended to emphasize geometric structure, junctions, and line connectivity, which are central for understanding circuit diagrams. Since circuit categories are often determined by component arrangement and connectivity rather than texture or color, edge representations provide a useful complementary view of the input.

3.4 Feature Extraction

We use a frozen `CLIP ViT-B/32` vision encoder. We implement feature extraction with `CLIPVisionModelWithProjection` for both RGB and edge inputs and keep the backbone frozen throughout training. This allows us to isolate the effect of representation-level fusion rather than full model adaptation. All extracted feature vectors are L2-normalized before downstream classification or fusion.

4 Experimental Setup

Our experiments compare zero-shot CLIP, supervised frozen-feature baselines, and our proposed SARF-based fusion methods.

4.1 Zero-Shot CLIP

For zero-shot evaluation, we follow the standard CLIP text-image matching setup. We build one prompt per class using the template:

```
"a circuit diagram of {class
name}"
```

Text prompts are encoded with CLIP's text encoder, image embeddings are extracted from the test images, and prediction is made by selecting the class with maximum cosine similarity between image and text embeddings. This experiment measures how well CLIP transfers to

the circuit domain without any task-specific supervised learning.

4.2 Supervised Classifiers

For supervised experiments, we extract frozen CLIP image features and train lightweight classifiers on top of them. For the linear models, we use multinomial logistic regression with `lbfgs` optimization, `max_iter=5000`, and balanced class weights. The regularization parameter C is selected using validation macro-F1 from the set:

$$\{0.01, 0.1, 1.0, 3.0, 10.0, 30.0, 100.0\}.$$

After selecting the best C on the validation set, the classifier is retrained on the combined training and validation features and evaluated on the test set.

For learned fusion, we use a two-hidden-layer MLP with hidden sizes 512 and 256, batch normalization, ReLU activations, and dropout. Early stopping is applied based on validation macro-F1 in order to reduce overfitting.

4.3 Evaluation Metrics

We report the following metrics:

- **Accuracy**
- **Macro-F1**
- **Top-5 Accuracy**
- **Confusion Matrix analysis**

Accuracy measures overall classification correctness. Macro-F1 is especially important in this setting because it gives equal weight to all classes and is more informative when some classes are much harder than others. Top-5 accuracy is also useful because the correct class may still appear among the model's highest-confidence predictions even when the top prediction is incorrect.

5 Baselines

Our baseline choices are motivated by the need to compare strong but standard frozen-backbone alternatives before evaluating our proposed fusion methods.

Zero-Shot CLIP. This is the most direct baseline and tests whether CLIP's pretrained alignment alone is sufficient for circuit classification. Its key assumption is that the text prompt space and image embedding space already transfer well to technical diagrams. In this domain, that assumption is weak because circuit diagrams differ significantly from the natural images used in CLIP pretraining.

5.2 Linear Probe (RGB-only). We freeze CLIP and train a linear classifier on F_{RGB}. This baseline measures how far frozen RGB CLIP embeddings can go with a simple supervised decision boundary.

5.3 Structural-Only Probe (Edge-only). We train a classifier on F_{Edge} alone to measure the standalone value of structural embeddings. This baseline isolates topology-focused information without relying on full RGB appearance cues.

Overall, these baselines fit the project context well because they provide meaningful points of comparison while preserving the same frozen CLIP backbone.

6 Proposed SARF Methods

Beyond the standard baselines, we evaluate our proposed **Structure-Aware Representation Fusion (SARF)** methods. These are **our experiments and proposed solutions**, not baselines. Their purpose is to test whether combining RGB and edge-derived structural features improves transfer performance in low-resource circuit classification.

SARF-Sum. In this method, RGB and edge embeddings are combined through element-wise

addition, followed by L2 normalization and logistic regression. This is the simplest fusion method and assumes that the two embedding spaces are sufficiently aligned for direct addition to be meaningful.

SARF-Concat. In this method, RGB and edge embeddings are concatenated into a larger feature vector and passed to logistic regression. This preserves the full information from both branches and allows the classifier to learn how much weight to place on each representation.

SARF-Learned Fusion. In this method, the concatenated RGB and edge features are passed through a multilayer perceptron. This allows nonlinear interactions between appearance based and structure based features,and also introduces more trainable parameters.

SARF-Ensemble. Finally, we evaluate an ensemble that averages the predicted probabilities from SARF-Sum, SARF-Concat, and SARF-Learned Fusion. This method is based on the idea that different fusion strategies may make different errors, and combining them may improve robustness.

7 Results

Table 1 summarizes the main results on the LITE circuit classification task. We explicitly separate standard baselines from our proposed SARF methods to make the evaluation clearer.

The results show that zero-shot CLIP performs poorly, obtaining only 5.84% accuracy and 0.0249 Macro-F1. This confirms that direct transfer from CLIP is not sufficient for this specialized low-resource domain.

Among the supervised baselines, RGB-only provides a large improvement over zero-shot performance, reaching 19.57% accuracy and 0.0789 Macro-F1. This suggests that CLIP visual features still contain useful information, but that they require task-specific supervision to become effective.

The Edge-only baseline performs worse than RGB-only, with 13.45% accuracy and 0.0552 Macro-F1. This indicates that structural information alone is helpful but insufficient by itself. Circuit edges capture topology and layout, but some semantic distinctions still benefit from the full RGB-based representation.

Among our proposed SARF methods, SARF-Concat is the strongest single model, reaching 19.94% accuracy and 0.0807 Macro-F1. SARF-Sum reaches the same accuracy but slightly lower Macro-F1. These results suggest that RGB and edge embeddings provide complementary signals, and that explicitly preserving both feature vectors through concatenation is somewhat more effective than forcing them into a single summed representation.

The SARF-Learned Fusion MLP underperforms the simpler linear methods, with only 8.07% accuracy and 0.0530 Macro-F1.

Finally, the SARF-Ensemble achieves the best overall Macro-F1 of 0.0832, although its accuracy (19.48%) is slightly below the best single-model accuracy.

8 Analysis

These results support two main conclusions. First, the LITE circuit classification task is genuinely difficult for standard CLIP transfer. The large gap between zero-shot CLIP and the supervised baselines shows that prompt-based transfer alone is not enough for technical diagrams.

Second, **structure-aware fusion helps, but modestly**. Edge information does not outperform RGB features alone, but it becomes more useful when combined with RGB embeddings. SARF-Concat is the strongest *single-model* variant in our results, with the best Top-1 accuracy and Macro-F1 among individual models. At the same time, SARF-Ensemble achieves the best overall Macro-F1, indicating a robustness benefit from combining multiple fusion strategies. Overall, structural priors appear complementary rather than sufficient on their own.

Model	Test Accuracy(Top-1 Accuracy)	Test Macro-F1	Test Top-5
Baselines			
Zero-Shot CLIP	0.0584	0.0249	–
RGB-only	0.1957	0.0789	0.4527
Edge-only	0.1345	0.0552	0.3683
Proposed SARF Methods			
SARF-Sum	0.1994	0.0783	0.4378
SARF-Concat	0.1994	0.0807	0.4258
SARF-Learned Fusion MLP	0.0807	0.0530	0.2542
SARF-Ensemble (Sum+Concat+MLP)	0.1948	0.0832	0.4267

Table 1: Comparison of standard baselines and proposed SARF methods on the LITE circuit diagram classification task. The SARF ensemble achieves the best Macro-F1, while SARF-Concat is the strongest single SARF model by Macro-F1.

8.1 Confusion Matrix Analysis (SARF-Concat)

Visualizing top confusion examples for SARF-Concat highlights recurring class-level errors:

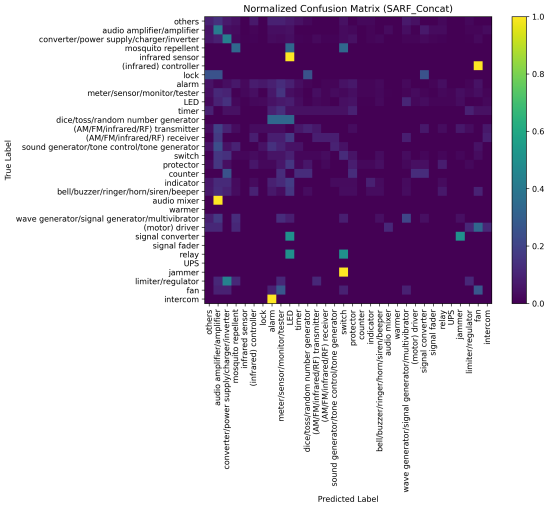

Figure 1: Confusion matrix for SARF-Concat on the LITE circuit classification task.

The confusion matrix is also consistent with this trend: many errors cluster among semantically and visually related classes, especially between *LED*, *converter/power supply/charger/inverter*, *audio amplifier/amplifier*, and *meter/sensor/monitor/tester*. This indicates that while SARF-Concat improves overall discrimination, fine-grained boundaries between functionally similar circuit categories remain a key challenge in LITE.

9 Baselines and Assumptions

We compare our approach against CLIP-based baselines reported in prior work on circuit diagram classification.

Method	Top-1 (%)	Top-5 (%)
Zero-Shot CLIP	7.7	28.5
Paper Baseline (CLIP)	19.4	45.3

Table 2: CLIP-based baselines for circuit diagram classification.

Zero-Shot CLIP. This baseline assumes that pretrained image-text alignment generalizes to circuit diagrams without training. However, this assumption does not hold well due to strong domain shift and the structural nature of circuits, resulting in low performance (7.7%). This baseline assumes that CLIP visual features are transferable and can be adapted using a simple classifier. It achieves the best baseline performance (19.4%), but still does not explicitly model structural properties such as connectivity and topology.

Summary. Overall, these baselines show that CLIP features are useful but insufficient for circuit understanding. Their assumptions only partially hold, motivating approaches that incorporate structural information explicitly.

10 References

References

[1] Yunhua Zhang, Hazel Doughty, and Cees G. M. Snoek. Low-resource vision challenges for foundation models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024.